DE LA SALLE HOLY CROSS COLLEGE

Grade 10 **Date: 3 November 2021**

Physical Sciences **Total: 130 marks**

Examination Paper 1: Physics **Time: 2 hours**

Examiner: A. Kamedza **Moderator: K. Tam**

INSTRUCTIONS AND INFORMATION

1. This question paper consists of 11 pages and SIX questions.
2. Answer ALL the questions on folio paper provided except Questions 1, 2.1.7 and 2.1.8 to be done on the printed part of your answer sheet.
3. It is in your own interest to write legibly and present your work neatly.
4. Number the answers correctly according to the numbering system used in the question paper.
5. Leave ONE line open between two subsections, for example between QUESTION 2.1 and QUESTION 2.2.
6. You may use a non-programmable calculator.
7. You may use appropriate mathematical instruments.
8. You are advised to use the attached DATA SHEETS.
9. Show ALL formulae and substitutions in ALL calculations.
10. Round off your final numerical answers to a minimum of TWO decimal places.

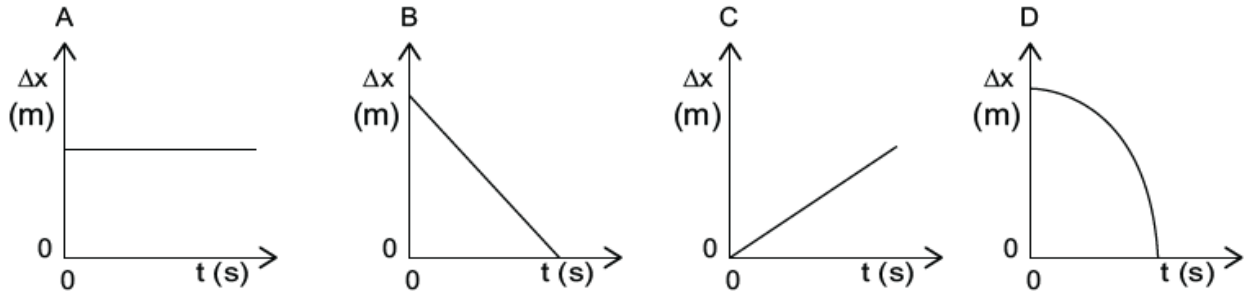
QUESTION 1 MULTIPLE CHOICE

Choose the correct answer and write the appropriate letter (A-D) next to the question number on your answer sheet, e.g. 1.1 E

1.1 Which one of the following pairs of quantities has one scalar and one vector quantity?

A distance and speed
B displacement and velocity
C acceleration and time
D force and acceleration

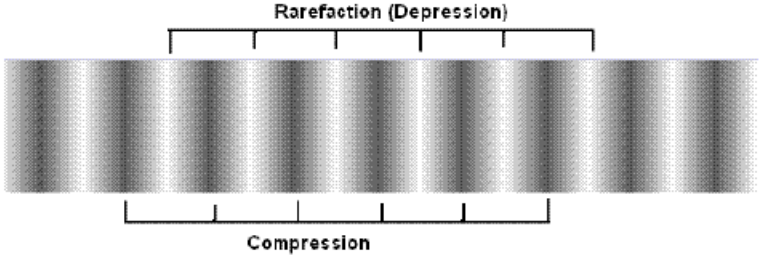
1.2 Which graph correctly shows the variation of position versus time for an object moving with constant acceleration?



1.3 A stone falls through the air from a high bridge. As it falls:

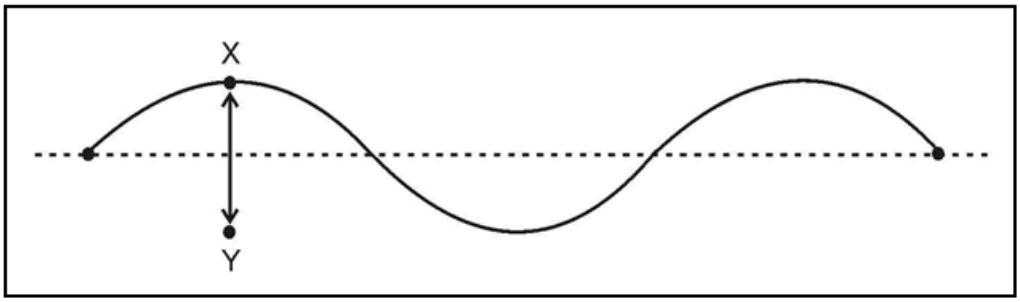
A mechanical energy is conserved.
B kinetic energy is conserved.
C potential energy is conserved.
D potential energy is transferred to kinetic energy and heat.

1.4 Compressions and rarefactions are characteristic of …



A transverse waves.
B longitudinal waves.
C both of these.
D neither of these.

1.5 Consider the diagram below of a wave with a frequency of 5 Hz.



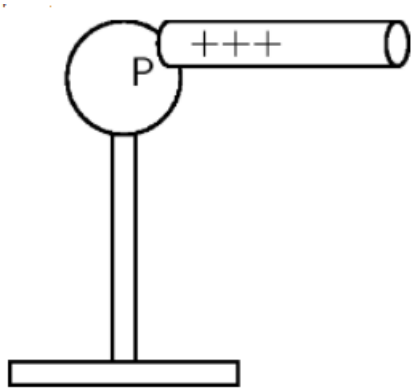
The distance XY can be described as:

A the wavelength.
B the amplitude.
C the period.
D twice the amplitude.

1.6 Which one of the following materials is not ferromagnetic?

A iron
B nickel
C copper
D cobalt

1.7 An uncharged hollow metal sphere is placed on an insulated stand. A positively charged rod is brought up to touch the hollow metal sphere at P as shown in the diagram below. It is then moved away.



Where is the excess charge distributed on the sphere after the rod has been removed?

A It is still located at point P where the rod touched the sphere.
B It is evenly distributed over the outer surface of the hollow sphere.
C It is evenly distributed over the outer and inner surfaces of the hollow sphere.
D No charge remains on the hollow sphere.

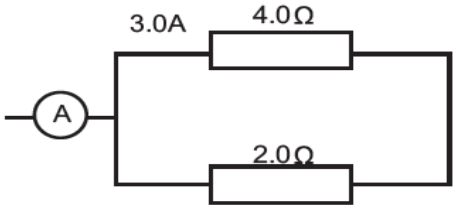
1.8 When a charged object becomes polarized, it …

A gains electrons.
B loses electrons.
C gains a north and a south pole.
D experiences separation of charge.

1.9 How much energy is transferred by a 6 Ω resistor in 10 seconds when 2,0 A of current passes through it?

A 2,4 J
B 14,4 J
C 240 J
D 1440 J

1.10 The diagram below shows part of an electrical circuit.



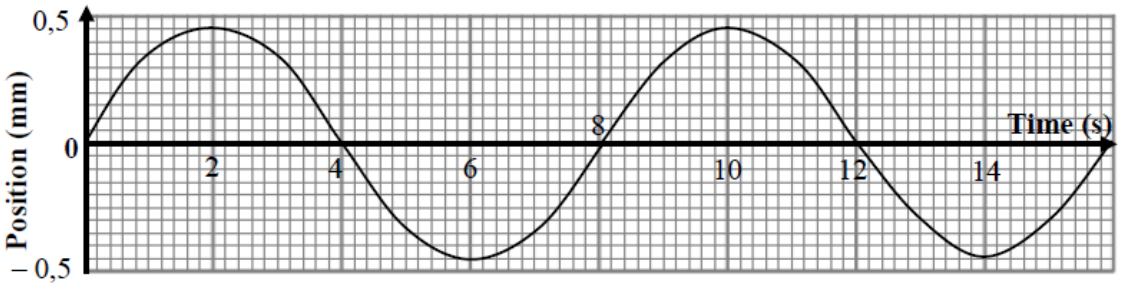
The current in the 4,0 Ω resistor is 3,0 A.
What is the reading on the ammeter?

A 4,5 A
B 6,0 A
C 9,0 A
D 12,0 A

[10 x 2 = 20]

QUESTION 2 WAVES

2.1 The position-time graph for a particle of the medium through which a sound wave moves is given below:



2.1.1 What type of wave is a sound wave? (1)

2.1.2 Explain how the wave, mentioned in Question 2.1.1, is propagated. (2)

2.1.3 What is the amplitude of the wave in the graph above? (1)

2.1.4 Calculate the frequency of the source through which the above sound wave is generated. (3)

2.1.5 Determine the period of the wave. (2)

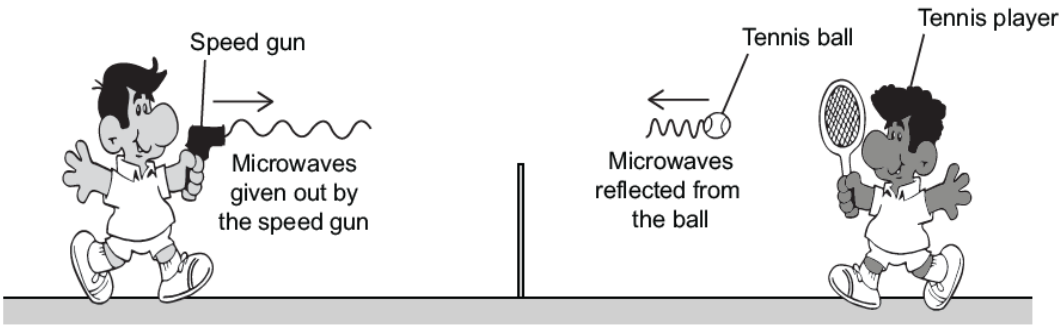
2.1.6 Calculate the wavelength of this wave if its speed is 0,15 $m \cdot s^{-1}$. (3)

On the **ANSWER SHEET** provided complete Questions 2.1.7 and 2.1.8

2.1.7 The frequency of the source that generates the above wave is suddenly doubled. Draw a sketch graph of position versus time for the same particle. Indicate the displacement values as well as values for the first 8 seconds. (4)

2.1.8 On this graph, indicate 2 points that are in phase and label them **A** and **B**. (1)

2.2 The diagram below shows a tennis coach using a "speed gun" to measure how fast the player serves the ball.



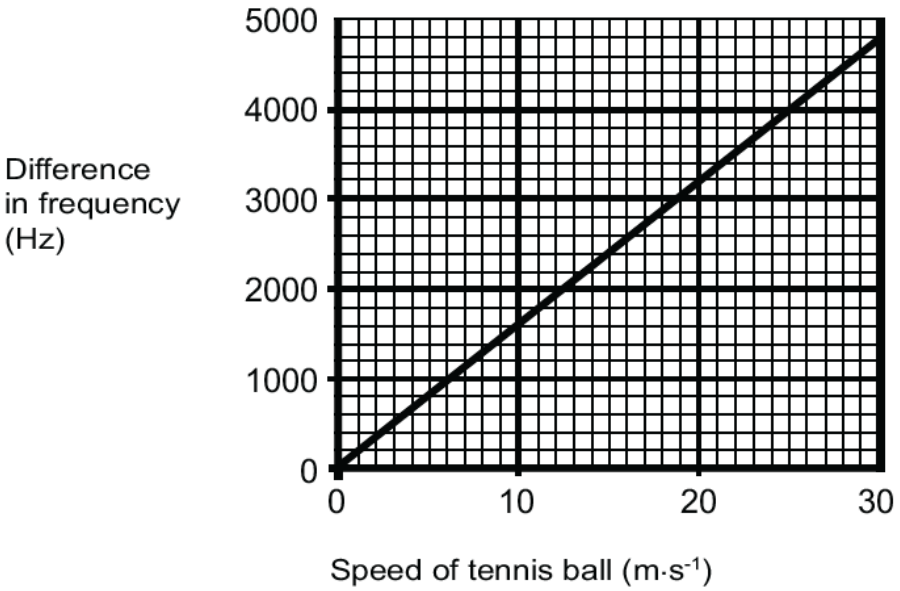
The microwaves transmitted by the speed gun have a frequency of 24 x 10^9 Hz.

2.2.1 Calculate the wavelength of the microwaves emitted from the speed gun. (3)

2.2.2 Some of the microwaves emitted are absorbed by the tennis ball. What effect will these absorbed waves have on the tennis ball? (1)

2.2.3 Some of the microwaves emitted are reflected by the moving tennis ball. There is an apparent change in the frequency of the microwaves when they are received by the speed gun. The speed gun calculates the difference in the frequency and using this information, it can estimate the speed of the tennis ball. The graph shown below, shows the relationship between the difference in frequency and the speed of the tennis ball.

Graph of difference in frequency against speed of the tennis ball



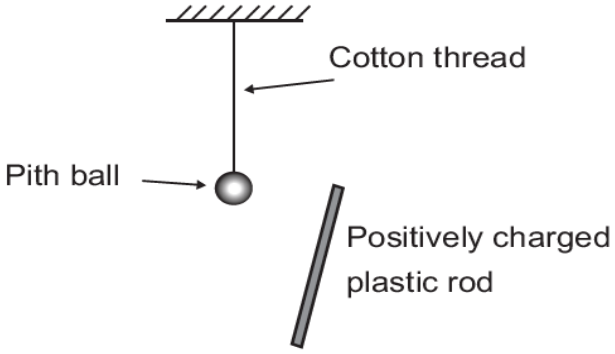
Describe the relationship between the difference in frequency and the speed of the tennis ball. (2)

2.2.4 When the difference in frequency is 3 200 Hz, what is the speed of the tennis ball? (2)

[25 marks]

QUESTION 3 ELECTROSTATICS

A pith ball is a polystyrene sphere coated with metal paint.
A plastic rod is charged by rubbing it with a cloth. It is then held next to an uncharged pith ball that is suspended on a light cotton thread.



3.1 Describe how the plastic rod becomes positively charged when it is rubbed with a cloth. (2)

3.2 Describe what happens in the metal paint on the pith ball when the positively charged plastic rod is brought near to it. (2)

3.3 The ball is attracted to the rod. Explain why this happens, given that the pith ball is uncharged. (2)

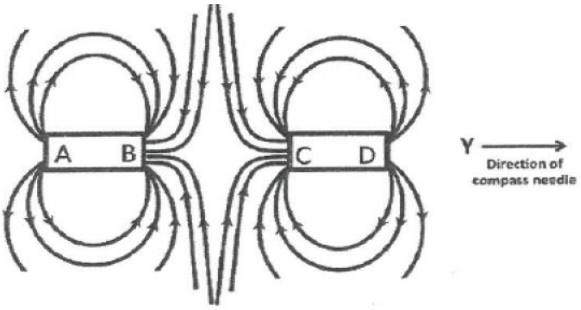
3.4 Predict what you would see if the pith ball touches the positively charged rod. (2)

3.5 Explain your answer to Question 3.4. (2)

[10 marks]

QUESTION 4 MAGNETISM

The diagram below shows the magnetic field pattern of two identical bar magnets with ends AB and CD.



4.1 Define the term *magnetic field*. (2)

4.2 State 2 characteristics of magnetic field lines. (2)

4.3 What important information about the polarities of B and C can be obtained from the diagram? (2)

A small compass is placed at point Y. The north pole of the compass points away from D.

4.4 Do the field lines inside the magnet CD point from **C to D**, or from **D to C**? Give a reason for your answer. (3)

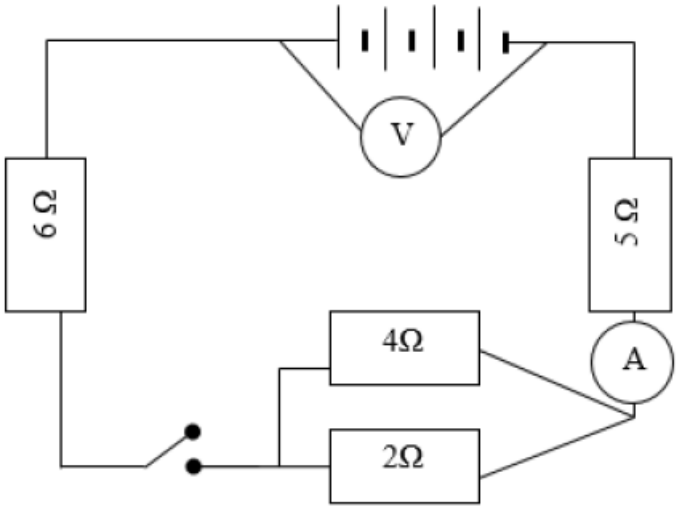
4.5 If magnet AB and CD were moved closer to each other, what effect will this have on the magnetic force between them? Explain your answer. (2)

4.6 Give a reason why the Earth's magnetic field is so important to our existence. (1)

[12 marks]

QUESTION 5 ELECTRIC CIRCUITS

Use the circuit diagram below to answer the questions that follow.
The voltmeter V connected across the cells reads 10 V. The switch in the circuit is closed and the current starts to flow.



5.1 Define the term *potential difference*. (1)

5.2 When **the switch is open**; determine the readings on the following meters:

5.2.1 voltmeter (1)

5.2.2 ammeter (1)

The switch in the circuit is now closed and current starts to flow.

5.3 Calculate the total effective resistance of the circuit. (3)

5.4 Determine the reading on the ammeter A. (2)

5.5 Calculate the potential difference across the 4 Ω resistor. (3)

5.6 Calculate the potential difference across the 6 Ω resistor. (2)

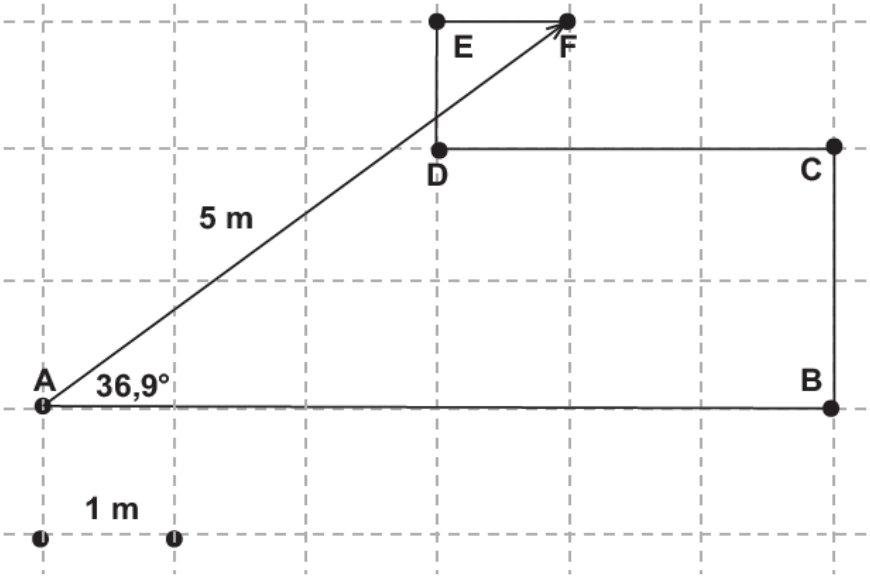
5.7 Determine the current through the 4 Ω resistor. (3)

5.8 Will the current through the 2 Ω resistor be **BIGGER** or **SMALLER** than the current calculated in Question 5.7? Explain. (2)

[18 marks]

QUESTION 6 MECHANICS

6.1 An object moves from A to B to C to D to E to F in 30 seconds, as shown in the diagram below:



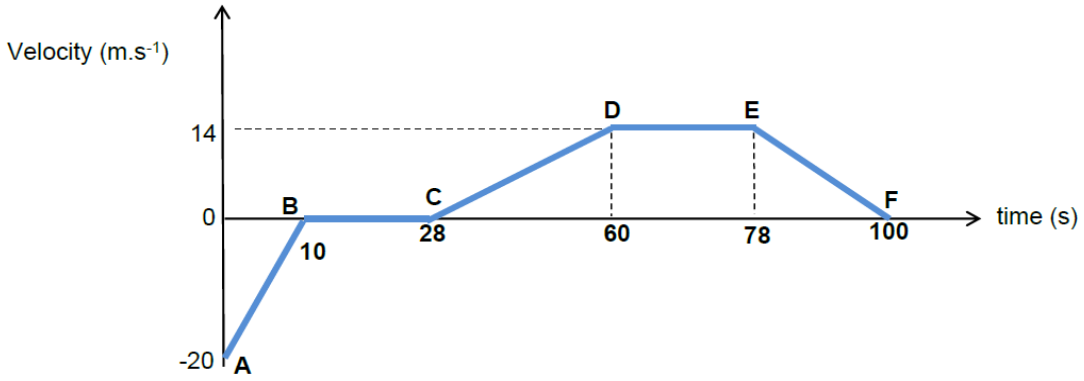
6.1.1 Calculate the distance moved in 30 s. (2)

6.1.2 The vector drawn from A to F shows the object's displacement in 30 s. With reference to the definition of displacement, explain why this vector gives the object's displacement, and write down the magnitude and direction of its displacement. (3)

6.1.3 Calculate the average speed of the object in 30 s. (3)

6.1.4 Calculate the magnitude of the object's average velocity in 30 s. (3)

6.2 Consider the velocity-time graph of a car travelling along a straight horizontal road. The diagram is not drawn to scale, North is the positive direction.



6.2.1 Define *acceleration*. (2)

6.2.2 Fully describe the motion of the car for interval AB (0 -10 s). (2)

6.2.3 Calculate the acceleration of the car for the interval EF. (3)

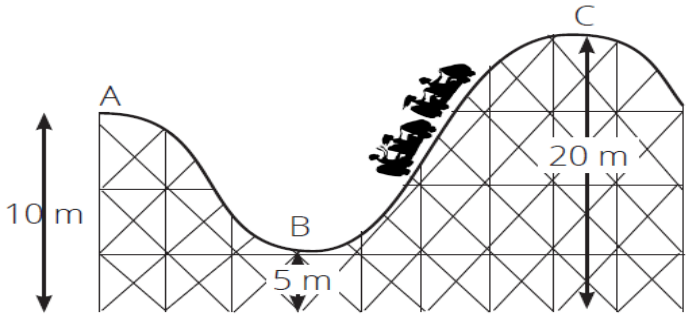
6.2.4 Determine the resultant displacement of the car. (4)

6.2.5 What is the magnitude of the average velocity of the car during the first minute? (3)

6.2.6 Use the letters A, B, C, D, E and F from the graph and write down the letter that corresponds to the time when the car was the furthest South. (1)

6.2.7 Sketch a position vs. time graph of the motion of the car. Take the position at the start (t=0) as the origin (reference point). (6)

6.3 The diagram below shows a roller coaster. The cart passes point **A** at a speed of 6 m·s^{-1}. **A** is 10 m high, **B** is 5 m and **C** is 20 m high. Consider the cart an isolated system and the frictional force of the track can be ignored.



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6.3.1 Define the term *mechanical energy*. (1)

6.3.2 How does the mechanical energy of the cart at **A** compare to that at **B**? Explain your answer. (2)

6.3.3 If the cart and its passengers together have a mass of 500 kg, calculate the total mechanical energy of the cart and passengers. (3)

6.3.4 Calculate the speed of the cart at point **B**. (4)

6.3.5 Will the cart have enough energy to reach point **C** without the help of an engine? Motivate your answer. (3)

[45 marks]

Total: 130 marks

GRADE 10 PHYSICAL SCIENCE PAPER 1 ANSWER SHEET

NAME: **DATE:**

QUESTION	OUT OF	MARK
1	20	
2	25	
3	10	
4	12	
5	18	
6	45	
TOTAL	130	

QUESTION 1

1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10

QUESTION 2.1.7 & 2.1.8

